

DSAA Lab 05

Queues

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CT-159: Data Structure Algorithms and Applications (Pr)

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Exercise

Q1: Write a Java program to reverse the elements of a queue.

Answer:

```
import java.util.*;

public class ReverseQueue {
    public static void main(String[] args) {
        Queue<Integer> q = new LinkedList<>();
        q.add(10);
        q.add(20);
        q.add(30);
        q.add(40);

        System.out.println("Original Queue: " + q);

        Stack<Integer> stack = new Stack<>();
        while (!q.isEmpty()) {
            stack.push(q.remove());
        }
        while (!stack.isEmpty()) {
            q.add(stack.pop());
        }

        System.out.println("Reversed Queue: " + q);
    }
}
```

Q2: Write a Java program to find the maximum element of a queue

Answer:

```
import java.util.*;

public class MaxInQueue {
    public static void main(String[] args) {
        Queue<Integer> q = new LinkedList<>();
        q.add(5);
        q.add(9);
        q.add(2);
        q.add(7);

        System.out.println("Queue: " + q);

        int max = q.peek();
        for (int x : q) {
            if (x > max) max = x;
        }

        System.out.println("Maximum element: " + max);
    }
}
```

```
}
```

Q3: Write a Java program to find the second highest element of a Queue.

Answer:

```
import java.util.*;

public class SecondHighestInQueue {
    public static void main(String[] args) {
        Queue<Integer> q = new LinkedList<>();
        q.add(5);
        q.add(9);
        q.add(2);
        q.add(7);

        System.out.println("Queue: " + q);

        int max = Integer.MIN_VALUE;
        int second = Integer.MIN_VALUE;

        for (int x : q) {
            if (x > max) {
                second = max;
                max = x;
            } else if (x > second && x != max) {
                second = x;
            }
        }

        System.out.println("Second highest element: " + second);
    }
}
```

Q4: Write a Java program to find the second lowest element of a queue

Answer:

```
import java.util.*;

public class SecondLowestInQueue {
    public static void main(String[] args) {
        Queue<Integer> q = new LinkedList<>();
        q.add(8);
        q.add(3);
        q.add(5);
        q.add(1);

        System.out.println("Queue: " + q);

        int min = Integer.MAX_VALUE;
        int second = Integer.MAX_VALUE;

        for (int x : q) {
            if (x < min) {
                second = min;
                min = x;
            } else if (x < second && x != min) {
                second = x;
            }
        }

        System.out.println("Second lowest element: " + second);
    }
}
```

Q5: Write a Java program to remove all the elements from a queue.

Answer:

```
import java.util.*;

public class RemoveAllFromQueue {
    public static void main(String[] args) {
        Queue<Integer> q = new LinkedList<>();
        q.add(10);
        q.add(20);
        q.add(30);

        System.out.println("Original Queue: " + q);

        q.clear(); // removes all elements

        System.out.println("Queue after removing all elements: " + q);
    }
}
```

